

Rajvardhan Patil

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EDUCATION

University of Colorado, Boulder (CU Boulder)

Master's in computer science

Coursework: Quantum computing, Database systems, Data mining, Neural Networks, Deep Learning, Big Data Architecture.

Colorado, U.S.A.

August 2025 - May 2027

Vellore Institute of Technology, Vellore (VIT)

Bachelor's in computer science and engineering; GPA: 3.4

Coursework: Data structure algorithms, Theory of computation, Design and analysis of algorithms, Software engineering

Tamil Nadu, INDIA

July 2021 - May 2025

EXPERIENCE

University of Colorado (CEAE Department)

Software Engineering Student Assistant

Boulder, CO, USA

March 2026 - Present

- Architected and developed a full-stack web application for building energy audit analysis, converting a PyQt5 desktop tool into a browser-accessible platform serving engineers and energy consultants across multiple commercial building projects.
- Benchmarked UI/UX patterns from industry tools (KAPSARC KBEAT2, UC Berkeley CBE Thermal Comfort Tool) to design an intuitive multi-step workflow covering building property input, utility data upload, weather integration, and EEM results visualization.
- Engineered a RESTful API backend using FastAPI wrapping a Python energy modeling engine (change-point inverse models, Building Load Coefficient analysis, EEM savings calculations), enabling asynchronous execution of long-running NOAA weather fetch and EEM analysis jobs.
- Delivered a React + TypeScript frontend with Plotly-powered interactive charts (psychrometric, end-use breakdown, measure ROI rankings), replacing 78-field PyQt5 forms with a tabbed, validated web UI.

Walstar Technologies

Software Engineering Intern

Kolhapur, MH, India

August 2023 - October 2023

- Engineered end-to-end development of a rural dairy automation app used by 5+ cooperatives, reducing manual operations by 40% and saving each cooperative approximately 7 hours per week.
- Architected and prototyped UI/UX in Figma, Flutter frontend for android with 95% UI test coverage; integrated with scalable Laravel backend.
- Achieved robust REST APIs using PHP and MySQL, achieving 99.8% uptime and preventing data loss during testing, which facilitated reliable data transmission for end users.

Valsco Technologies

Software Engineering Intern

Vellore, TN, India

May 2023 - July 2023

- Programmed and validated push notification workflows and in-app navigation flows, increasing feature adoption by 12%.
- Initiated to 4+ new features for the Jurident by implementing Flutter widgets, authentication workflows, and API integrations, increasing user session time by 18%.

VOLUNTEER

Computer Association of Kolhapur (CAK)

Software Developer

Kolhapur, MH, India

January 2026 - Present

- Designed and built a full-stack membership portal handling new registrations and renewals, replacing a manual process with a multi-step digital onboarding flow covering personal, business, and payment details.
- Integrated Razorpay payment gateway with automatic PDF receipt generation using ReportLab, triggering membership activation and expiry extension upon successful payment confirmation.
- Built a JWT-secured admin dashboard exposing real-time membership stats (active, pending, expired, revenue) and a bulk Excel import pipeline that parses, validates, and deduplicates member records before inserting them into the database.

Konark Computers

Software Engineering Intern

Kolhapur, MH, India

January 2025 - May 2025

- Collaborated a project to automate the resolution of 5 common networking issues through scripting, which boosted self-service adoption rates by reducing reliance on manual intervention.
- Implemented telemetry dashboards using Grafana for real-time monitoring of LAN/WAN performance, enabling data-driven optimization that improved application response times by 20%.
- Revamped the maintenance schedule for 12 high-volume printers, focusing on paper feed mechanisms and toner replacements, which cut printer-related service requests by 45%.

VinnovateIT

Application Developer Lead
Senior Core Member

Vellore, TN, India

December 2023 - December 2024
November 2022 - November 2023

- Teamed Bunkbuddies website functionality using React, and redesigned UI with Figma, improving user engagement by 35% and

reducing development time by 20% through reusable component design.

- Leveraged AI coding assistants (GitHub Copilot) to accelerate development velocity by 30% while maintaining code quality standards through structured code review processes.
- Implemented comprehensive end-to-end testing strategies for the MESSIT application and drove code maintenance initiatives, which improved code quality, leading to a 10% decrease in recurring bug reports.

PROJECTS

- **GradBro** ([link](#)): AI-powered Statement of Purpose (SOP) editor with ideation, writing assistance, and review features for college application essays. Supports 1500+ users. Tech: Next.js, React, LLMs, AWS CloudFront, and real-time text analysis.
- **Gradmits** ([link](#)): Complete graduate admissions toolkit providing personalized university recommendations, application tracking, and consulting services for US Master's programs. Tech: Next.js, React, Tailwind CSS, AWS, LLM's, FastAPI and MongoDB.
- **Saber** ([link](#)): Full-stack building energy audit platform that guides users through a multi-step audit workflow — geometry, envelope, HVAC, equipment, and ECM selection and runs change point analysis on utility data to generate energy conservation measure reports with PDF export. Tech: Next.js 14, React, TypeScript, Tailwind CSS, FastAPI, Python, Pandas, SciPy, Matplotlib, and REST API architecture.
- **CAK Membership Portal** ([link](#)): Built a full-stack membership management system for the Computer Association of Kolhapur, featuring multi-step onboarding with GST certificate and photo uploads to Supabase Storage, Razorpay payment integration with automated PDF receipt generation, and a JWT-secured admin dashboard with bulk Excel member import. Tech: Next.js, TypeScript, Tailwind CSS, FastAPI, PostgreSQL, SQLAlchemy, Supabase, Razorpay.
- **Pumped Up Kicks** ([link](#)): AI-powered lecture intelligence platform that converts video lectures into searchable, timestamped transcripts with conversational Q&A and instant navigation to relevant moments. Tech: React, TypeScript, FastAPI, Python, RAG, Vector Database (ChromaDB), LLMs (Ollama), Whisper STT, and microservice architecture.
- **Redbro** ([link](#)): Built Reddit marketing automation system for AI video SaaS, featuring high-intent post discovery engine with ML-based intent scoring, multi-account management with rate limiting, and automated engagement workflows. Tech: Python, PRAW, PostgreSQL, Redis, OpenAI API, Docker.
- **Sober-Space** ([link](#)): Developed anonymous substance addiction reporting with AI-generated insights via LLM-powered analysis and secure FastAPI backend. Tech: React, Next.js, Tailwind CSS, FastAPI, LLM.
- **Dret** ([link](#)): Incorporated canvas combined with hand gesture detection technology to enable intuitive, real-time drawing and interaction on the board in video conference web application. Tech: React, Next.js, MediaPipe, FastAPI, LiveKit, and TailwindCSS.
- **Messit** ([link](#)): Implemented and refined real-time push notifications for MessIt's menu updates, which directly contributed to a 10% increase in user engagement with daily menu options. Tech: React, Node.js, MongoDB, Redux, Firebase push notifications.

SKILLS

- **Frameworks:** Node.js, React.js, Next.js, Redux, Express.js, FastAPI, Laravel, Flutter, MongoDB, PostgreSQL, MySQL, GraphQL, Firebase, Tailwind CSS.
- **AI/ML:** scikit-learn, TensorFlow, RAG, LLMs (GPT, Claude, Ollama), Vector Databases, GitHub Copilot, Claude Code.
- **Tools:** Git, AWS (S3, CloudFront, Lambda), Docker, REST APIs, Linux, Postman, Figma, GitHub Actions, Vercel, Grafana, Cypress, Jest, LiveKit, Whisper STT, MediaPipe.
- **Skills:** Agile, SCRUM, SDLC, CI/CD, Microservices, gRPC, RESTful APIs, Telemetry & Observability, Real-time Analytics, UI/UX Design, Test-Driven Development.
- **Languages:** JavaScript (ES5/ES6), TypeScript, Python, Java, HTML, CSS, PHP.